Visual Data Analysis

UNIVERSITY OF SKÖVDE

Cost Inefficient Cities in the USA

Georgios Kokolakis

Visual Data Analysis report
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Introduction

Data visualization is a powerful tool for exploring and analysing complex information, and it is especially useful when it comes to exploring trends in economic data. This paper will focus on the analysis of living costs in different states in the United States of America. Using the visualization tool Tableau, we will explore the variations in the cost of living in different cities and the factors that contribute to these variations. The results of this analysis will answer our research questions, as well as provide valuable insights for individuals who are interested in understanding the cost of living in different cities and how these costs impact the standard of living for residents.

Background

For the purposes of acquiring a better understanding of visualization techniques the work of (Azzam, 2013) was used. Additionally, in order to measure the cost of living and formulate criteria to compare the respective cities I relied on (Moeller, n.d.). The first article provides an introduction to geographic information systems (GIS) and examples of how this method of data visualization can assist evaluators in better comprehending the environment in which they work, digging deeper into the data, and communicating by using maps to illustrate key findings and patterns. The authors also analyze this approach's drawbacks and make recommendations for those looking to incorporate GIS into their evaluation process. The second article offers insights to the Cost of Living Index (CoLI), published by the Council for Community and Economic Research (C2ER) every quarter.

The CoLI is divided into categories such as food, lodging, transportation, and health care. All values are determined for the index in proportion to the national average, which is equal to 100. The stated value for each metropolitan region is expressed as a percentage. For instance, an index of less than 100 indicates that the price of the good or service is lower than the national average, and a value of more than 100 indicates that the price is more than the average. By utilizing these indices, one is able to determine the most expensive cities in the country (Moeller, n.d.).

Having outlined an appropriate ranking system, we now turn to finding a suitable way to visualize the indices. GIS enables assessors to graphically analyse program outcomes and their distribution over a given region or geographic area. Depending on the goals of the evaluation, the resources available, and the evaluator's expertise, such images may be sophisticated or simple. The more intricate maps can be designed and customized to display a wide range of variables and results as well as to examine how results were influenced by environmental factors (Azzam, 2013). For instance, the Los Angeles Police

Department (LAPD) implemented the Safer Cities strategy in 2006, which led to the harsh punishment of minor infractions in downtown Los Angeles. With a focus on enforcement rather than prevention, this program sought to minimize the number of homeless people in the Skid Row neighbourhood. Cartifact[1] was hired by the LAPD in 2007 to map the effects of this policy after it was put into effect. A set of maps made by Cartifact showed how many homeless people there were in downtown Los Angeles. These maps vividly demonstrated how the Safer Cities strategy had reduced the number of homeless people in the target area over time, from November 2006 to June 2007. These maps also indicated where more resources were required, and the LAPD used them to interact with the general public. These analyses call for more advanced GIS and statistical knowledge, but they can yield crucial information regarding program performance over time.

Data

The dataset used is the Cost of Living Index for Selected U.S. Cities, 2010 (https://www2.census.gov/library/publications/2011/compendia/statab/131ed/tables/12s0728.xls) which provides the indices mentioned above (Groceries, Transportation, Utilities, Housing, Misc., CoLI) for selected US cities in 2010, and comes directly from the census department of the USA government. At the same time the dataset State and County Estimates for 2010 (https://www.census.gov/data/datasets/2010/demo/saipe/2010-state-and-county.html) , was used to contribute into a map layer for the average income in each state of the US in 2010.

One research question (or hypothesis)

Our research question is discovering the most expensive cities to live in the USA in 2010 utilizing the CoLI. At the same time, with the use of Average income by state we will manage to visualize the economically stronger and weaker states in the country based on their income and the CoLI of their cities.

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[1] Cartifact is US based company that creates impactful custom maps and graphics for commercial real estate, government, and technology firms (*Cartifact*, n.d.).

Visualization results

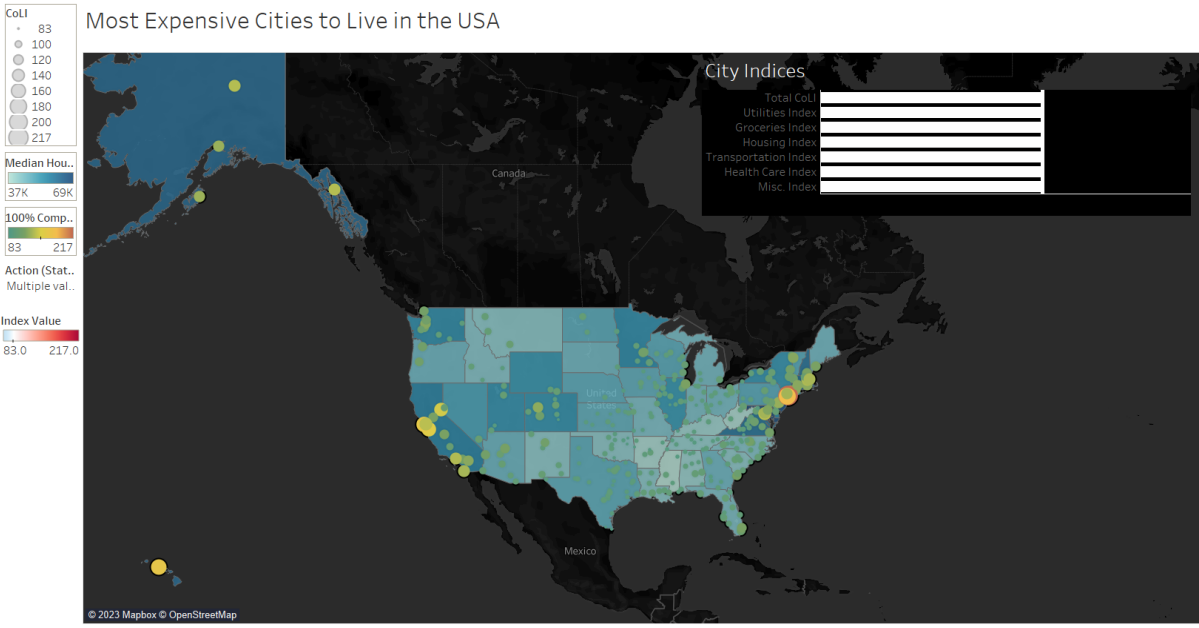

Figure 1 Cost of Living Index & Average Income by State in the USA

Data analysis

The answer to our research questions is given through a map chart. The map contains two layers. The first layer contains the information about the average income by state. The shades of the blue color are used to describe the states from wealthy (Dark Blue) to less wealthy (Light Blue).

The second layer contains information about the indices. Each city in the data is depicted with a circle mark on the map, the size and the color of the mark denotes the cities with the highest CoLI, i.e., the larger and redder the marker the higher the index. Green and yellow colors are used to describe low and average indices, respectively.

Furthermore, a bar chart is used in the top right corner of the map that gives insights about all the indices for a specific city. The white vertical line on the chart denotes the average value for the indices which is 100. Any bar of the chart that exceeds the vertical line would indicate that the specified city is averagely more expensive in the specified index, while any bar below the line indicates that the city is averagely cheaper. The colors of the bar chart range from shades of blue (below average index) to white (average index) to shades of red (above average index).

The map and the bar chart are interactive, meaning that clicking on any marker will highlight the selected city and State in which it belongs. A text

window will appear below the marker with the name of the City, the State, and the CoLI. Simultaneously, the bar chart will display the index values for the selected city (as shown in the following Example).

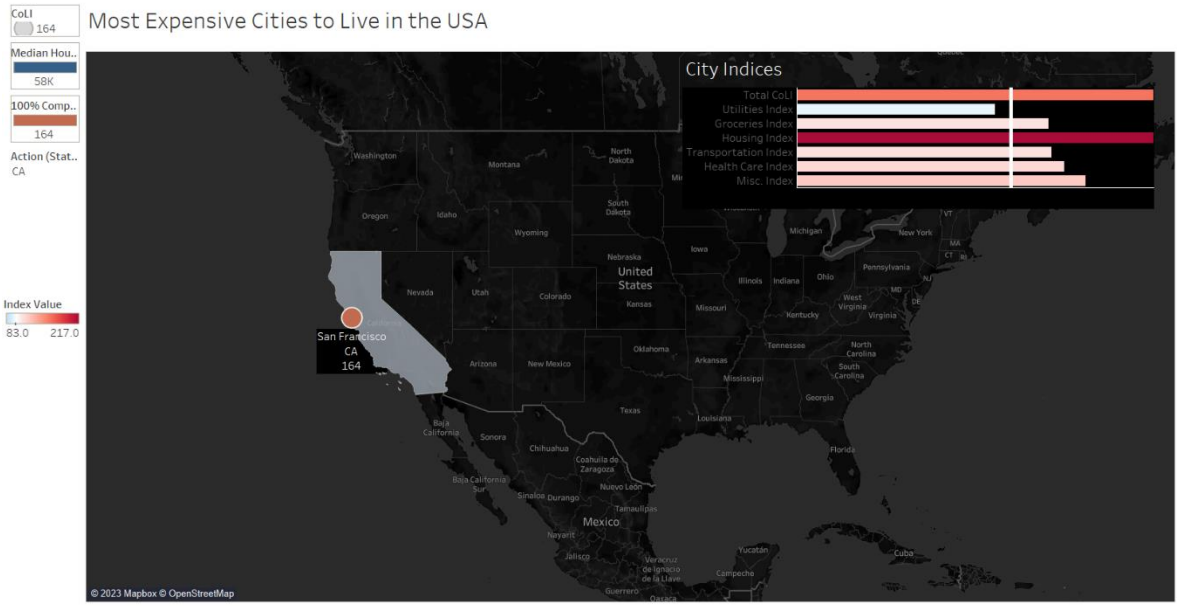

Figure 2 The Cost of Living in Queens NY

Moreover, the use of a spider chart helps the reader compare the four most expensive cities that originally stand out from the Map chart. The heptagon represents the average value of the indices (100) and it is clear that all four cities are more than exceeding the heptagon. The city of Harlingen is used as contrast, since according to our data it is the city with the smallest index value.

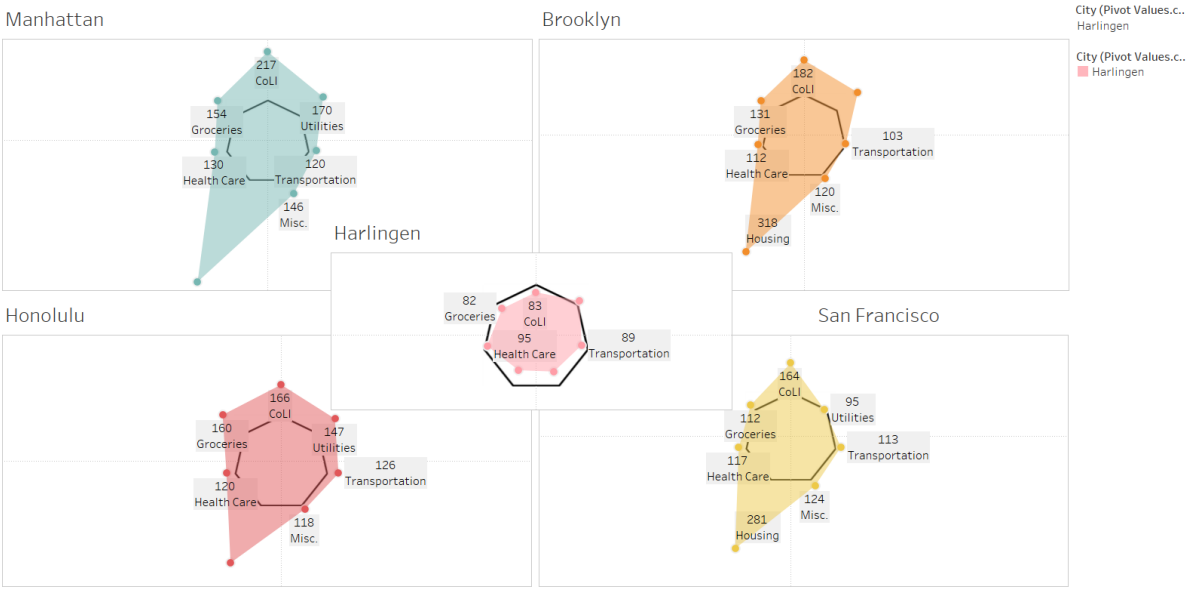

Figure 3 Spider Chart for the comparison of the most expensive cities

Reflection

The results of the visualization indicate that regarding the most expensive cities in the USA, the majority of them are located in California and New York. Manhattan NY appears to be the city with the highest cost of living with an index of 217, which means that Manhattan is more than twice as expensive to live in as the average city in the USA. The housing index alone in Manhattan is 387 making it the highest in the country, and pointing out that buying/renting a house in this city is averagely four times more expensive. Manhattan is followed by Brooklyn NY, Honolulu Hawaii, and San Francisco California. Brooklyn and Manhattan seem to follow the same pattern which is consistent considering the proximity of the cities. Honolulu scores the highest in groceries with an index of 160, which is also to be expected considering that Hawaii is an island thus the products consumed need to be imported from the mainland. Overall, housing plays a major part in the high cost of living index, as one can observe that the most expensive cities tend to surpass the 200 index threshold, which makes them more than twice as expensive in that category and con.

In regards to the average income by state question, Maryland seems to be the wealthiest state with 69 thousand dollars yearly on average, followed by New Jersey and Alaska. Interestingly enough, despite California and New York containing the cities with the highest cost of living, they are not the wealthiest States in the Country as both have more than a 10 thousand dollar yearly gap with Maryland.

By exploring the map, one is able to acquire various information on each city and answer a number of inquiries. According to one's needs and income, the map can be used to discover ideal cities for relocation. In our case the markers helped us understand the most expensive cities to live in considering all factors and showed the concentration of expensive cities in specific states in relation to the state's average income.

Given that the data available referred to locations, a GIS system was a suitable way to address the research questions, similarly with the LAPD example encountered in literature. Furthermore, the use of the spider chart made it possible to compare these cities side by side.

The visualization could be improved in given more time and resources. First and foremost, the dataset accounts for cities in 2010, thus being outdated. The Council for Community and Economic Research (C2ER) contains the latest datasets on the topic of CoLI in the USA, however it requires a subscription to download said datasets. Furthermore, given more time one could create similar visualizations for all of the indices, so as to determine the most expensive cities for Healthcare, Utilities etc.

While finding an answer to our research question challenges arose. An expected challenge is using the visualization tool as beginner in Visual Data Analysis. Although, tableau is a handy tool, one has to familiarize with it in order to produce the desired results. Additionally, another challenge was configuring the geolocation of the cities. Many cities have the same name, and representing them all in a map is not always trivial. Last but not least, one has to experiment with the different colors and types of visualization to make it easy to read and convey the message of the research question.

References

Azzam, T. (2013). Mapping Data, Geographic Information Systems. *New Directions for Evaluation*, *2013*(140), 69–84. https://doi.org/10.1002/ev.20074

Cartifact. (n.d.). Cartifact. Retrieved February 17, 2023, from https://cartifact.com

Moeller, C. (n.d.). *ACCRA cost of living index*. Retrieved February 15, 2023, from https://www.tandfonline.com/doi/epdf/10.1080/08963568.2020.1847558?needAccess=true&role=button

Appendix

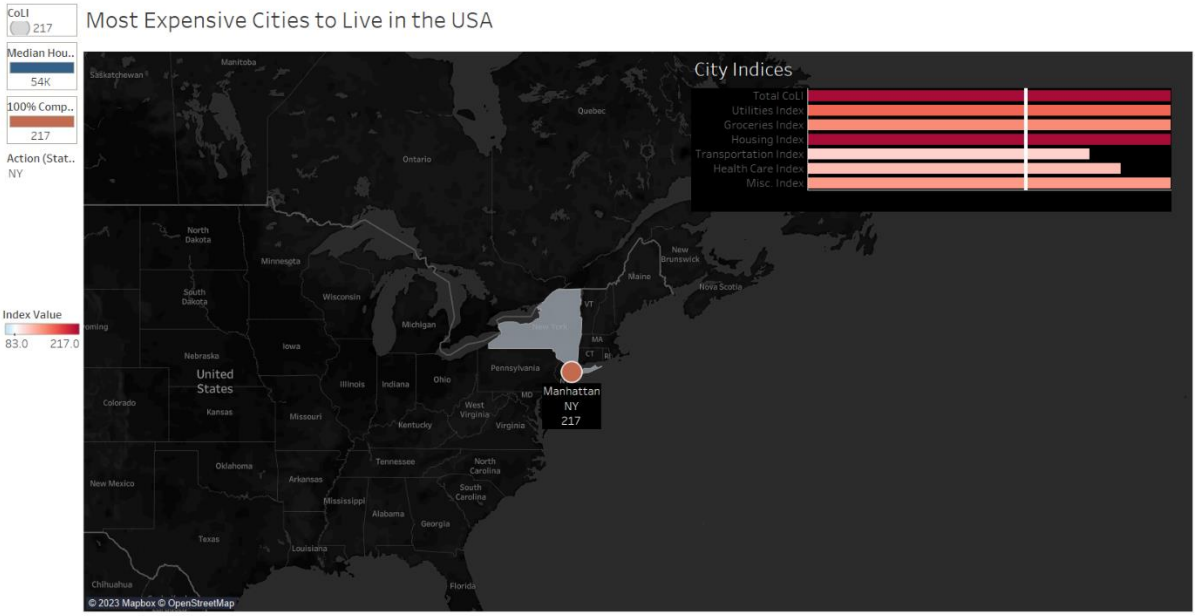

Figure 4 The cost of living in Manhattan NY

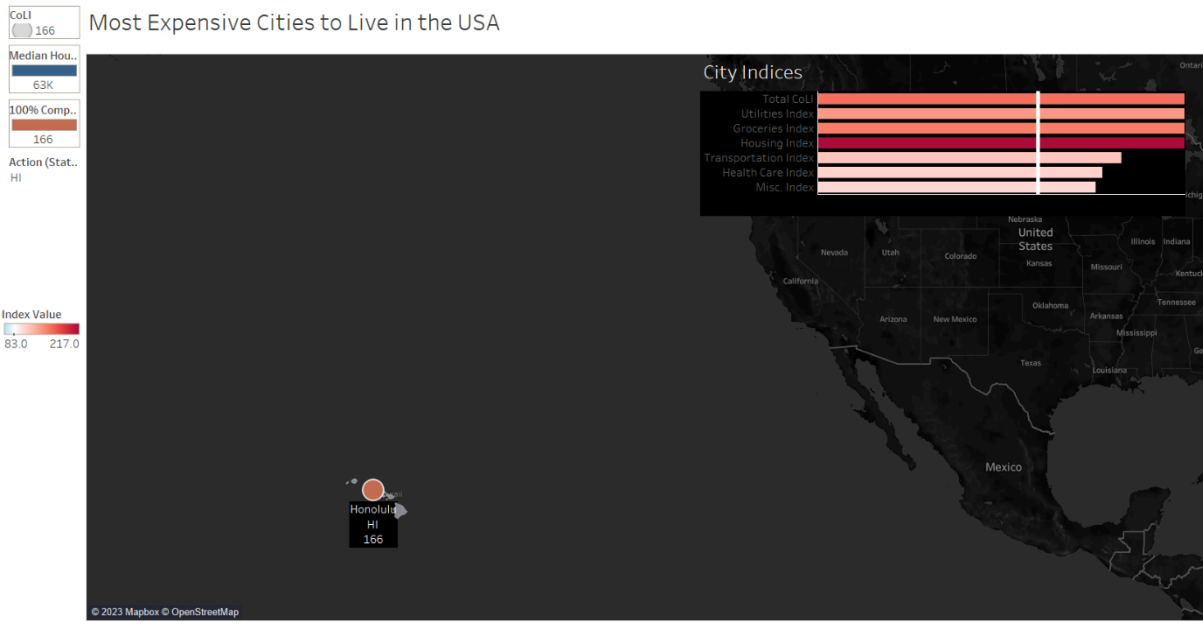

Figure 5 The cost of living in Honolulu HI

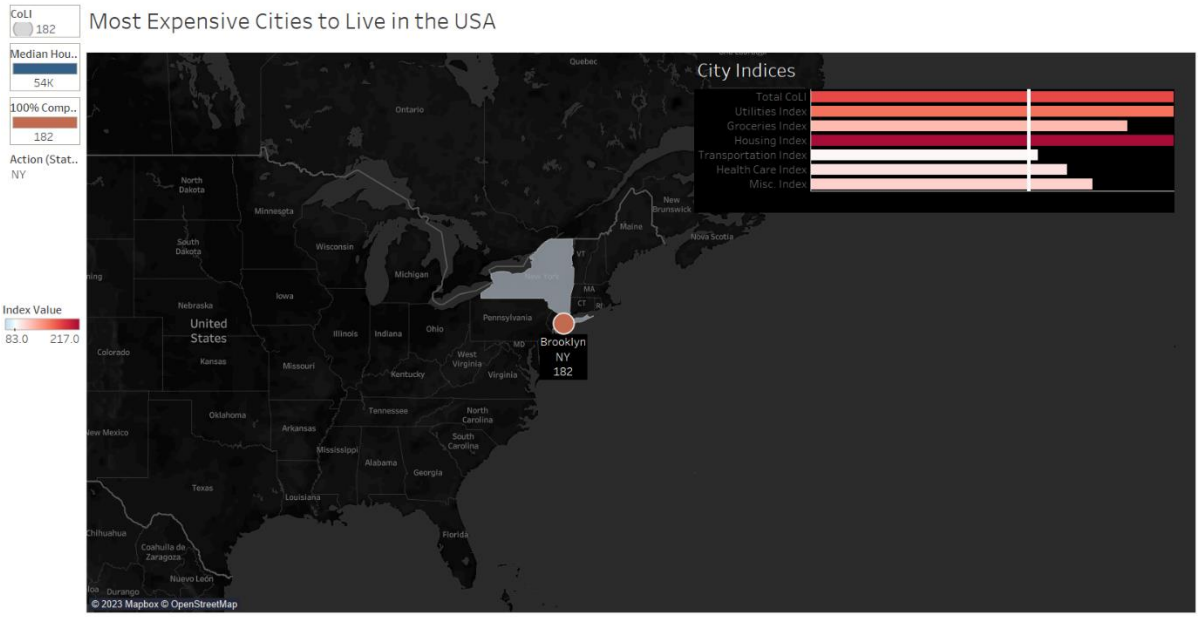

Figure 6 The cost of living in Brooklyn NY